$$= \frac{-2+4}{6} ; \frac{-2-4}{6}$$

$$= \frac{2}{6} ; \frac{-6}{6}$$

$x = 0.33 ; -1$ (not possible)

$\therefore [A_2]_{eq} = 1 - x = 1 - 0.33 = 0.67$

$[B_2]_{eq} = 1 - x = 1 - 0.33 = 0.67$

$[AB]_{eq} = 2x = 2 \times 0.33 = 0.66$

49. $K_{P_1} = 8.19 \times 10^2$ $T_1 = 298$ K

$K_{P_2} = 4.6 \times 10^{-1}$ $T_2 = 498$ K

$$\log\left(\frac{K_{P_2}}{K_{P_1}}\right) = \frac{\Delta H^\circ}{2.303\, R}\left(\frac{T_2 - T_1}{T_1 T_2}\right)$$

$$\log\left(\frac{4.6 \times 10^{-1}}{8.19 \times 10^2}\right) = \frac{\Delta H^\circ}{2.303 \times 8.314}\left(\frac{498 - 298}{498 \times 298}\right)$$

$$\frac{-3.2505 \times 2.303 \times 8.314 \times 498 \times 298}{200} = \Delta H^\circ$$

$\Delta H^\circ = -46181$ J mol^{-1}

$\Delta H^\circ = -46.18$ KJ mol^{-1}

50. $P_{CO_2} = 1.017 \times 10^{-3}$ atm $T = 500^\circ$ C

$K_P = P_{CO_2}$

$\therefore K_{P_1} = 1.017 \times 10^{-3}$ $T = 500 + 273 = 773$ K

$K_{P_2} = ?$ $T = 600 + 273 = 873$ K

$\Delta H^\circ = 181$ KJ mol^{-1}

$$\log\left(\frac{K_{P_2}}{K_{P_1}}\right) = \frac{\Delta H^\circ}{2.303\, R}\left(\frac{T_2 - T_1}{T_1 T_2}\right)$$

$$\log\left(\frac{K_{P_2}}{1.017 \times 10^{-3}}\right) = \frac{181 \times 10^3}{2.303 \times 8.314}\left(\frac{873 - 773}{873 \times 773}\right)$$

$$\log\left(\frac{K_{P_2}}{1.017 \times 10^{-3}}\right) = \frac{181 \times 10^3 \times 100}{2.303 \times 8.314 \times 873 \times 773}$$

$$\frac{K_{P_2}}{1.017 \times 10^{-3}} = \text{anti log of } (1.40)$$

$$\frac{K_{P_2}}{1.017 \times 10^{-3}} = 25.12$$

$$\Rightarrow K_{P_2} = 25.12 \times 1.017 \times 10^{-3}$$

$$K_{P_2} = 25.54 \times 10^{-3}$$

Unit - 9

Evaluate yourself :

1) mass of KOH = 5.6 g

no. of moles $= \frac{5.6}{56} = 0.1$ mol

i) Volume of the solution = 500 ml = 0.5 L

ii) Volume of the solution = 1 L

$$\text{molarity} = \frac{\text{number of moles of solute}}{\text{volume of solution (in L)}}$$

$$= \frac{0.1}{0.5} = 0.2 \text{ M}$$

ii) Volume of the solution = 1 L

molarity $= \frac{0.1}{1} = 0.1$M

2. mass of glucose = 2.82 g

no. of moles of glucose $= \frac{2.82}{180} = 0.016$

mass of water = 30 g

$$= \frac{30}{18} = 1.67$$

Key answers -jagan.indd 297 22-01-2021 13:23:21